

Basics of NLP

NLP - Natural Language Processing:

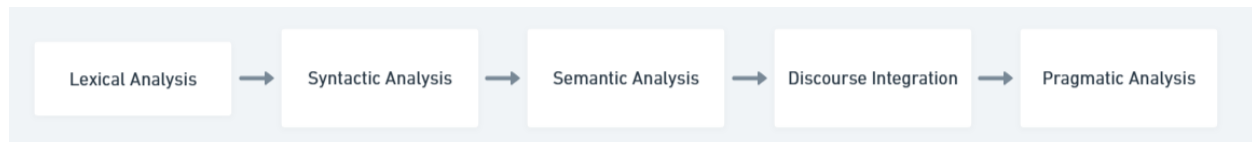
It helps machines to understand and process human languages either in text or audio form. It is used across a variety of applications from speech recognition to language translation and text summarization. It can be categorized into two components:

1. Natural Language Understanding: It involves interpreting the meaning of the text.
2. Natural Language Generation: It involves generating human-like text based on processed data.

Importance of NLP: Natural Language Processing (NLP) plays an important role in transforming how we interact with technology and understand data. Below are reasons why it's so important:

- Information Extraction: Extracts useful data from unstructured content.
- Sentiment Analysis: Analyzes customer opinions for businesses.
- Automation: Streamlines tasks like customer service and document processing.
- Language Translation: Breaks down language barriers with tools like Google Translate.
- Healthcare: Assists in analyzing medical records and research.

Phases of NLP:



Lexical and Morphological Analysis

Lexical Analysis: It focuses on identifying and processing words (or lexemes) in a text. It breaks down the input text into individual tokens that are meaningful units of language such as words or phrases. Key tasks in Lexical analysis:

- Tokenization: Process of dividing a text into smaller chunks called tokens. For example the sentence "I love programming" would be tokenized into ["I", "love", "programming"].
- Part-of-Speech Tagging: Assigning parts of speech such as noun, verb, adjective to each token in the sentence. This helps us to understand the grammatical roles of words in the context.

Example: Consider the sentence: "I am reading a book."

- Tokenization: Sentence is broken down into individual tokens or words: ["I", "am", "reading", "a", "book"]
- Part-of-Speech Tagging: Each token is assigned a part of speech: ["I" → Pronoun (PRP), "am" → Verb (VBP), "reading" → Verb (VBG), "a" → Article (DT), "book" → Noun (NN)]

Importance of Lexical Analysis

- Word Identification: It breaks text into tokens which helps the system to understand individual words for further processing.
- Text Simplification: It simplifies text through tokenization and stemming which improves accuracy in NLP tasks.

Morphological Analysis: It deals with morphemes which are the smallest units of meaning in a word. It is important for understanding the structure of words and their parts by identifying free morphemes (independent words like "cat") and bound morphemes (like prefixes or suffixes e.g. "un-" or "-ing"). Key tasks in morphological analysis:

- Stemming: Reducing words to their root form like "running" to "run".
- Lemmatization: Converting words to their base or dictionary form considering the context like "better" becomes "good".

Importance of Morphological Analysis

- Understanding Word Structure: It helps in breaking the composition of complex words.
- Improving Accuracy: It enhances accuracy of tasks such as part-of-speech tagging, syntactic parsing and machine translation.
- By identifying and analyzing morphemes the system can identify text correctly at the most basic level which helps in more advanced NLP applications.

Syntactic Analysis (Parsing): Syntactic Analysis helps in understanding how words in a sentence are arranged according to grammar rules. It ensures that the sentence follows correct grammar which makes the meaning clearer. The goal is to create a parse tree which is a diagram showing the structure of a sentence. It breaks the sentence into parts like the subject, verb and object and shows how these parts are connected. This helps machines understand the relationships between words in the sentence. Key components of syntactic analysis include:

- POS Tagging: Assigning parts of speech to words in a sentence as discussed earlier.
- Ambiguity Resolution: Handling words with multiple meanings ("book" can be a noun or verb).

Examples: Consider the following sentences:

- Correct Syntax: "John eats an apple."
- Incorrect Syntax: "Apple eats John an."

Despite using the same words, only the first sentence is grammatically correct and makes sense. The correct arrangement of words according to grammatical rules is what makes the sentence meaningful. By analyzing sentence structure NLP systems can better understand and generate human language. This helps in tasks like machine translation, sentiment analysis and information retrieval by making the text clearer and reducing confusion.

Semantic Analysis: Semantic Analysis focuses on understanding meaning behind words and sentences. It ensures that the text is not only grammatically correct but also logically coherent and contextually relevant. It aims to understand dictionary definitions of words and their usage in context and also find whether the arrangement of words in a sentence makes logical sense. Key Tasks in Semantic Analysis:

- Named Entity Recognition (NER): It identifies and classifies entities such as names of people, locations, organizations, dates and more. These entities provide important meaning in the text and help in understanding the context. For example in the sentence "Tesla announced its new electric vehicle in California," NER would identify "Tesla" as an organization and "California" as a location.
- Word Sense Disambiguation (WSD): Many words have multiple meanings depending on the context in which they are used. It identifies the correct meaning of a word based on its surrounding text. For example, the word "bank" can refer to a financial institution or the side of a river. It uses context to identify which meaning applies in a given sentence which ensures that interpretation is accurate.

Example of Semantic Analysis

"Apple eats a John." while grammatically correct this sentence doesn't make sense semantically because an apple cannot "eat" a person. Semantic analysis ensures that the meaning is logically sound and contextually appropriate. It is important for various NLP applications including machine translation, information retrieval and question answering.

Discourse Integration: It is the process of understanding how individual sentences or segments of text connect and relate to each other within a broader context. This phase ensures that the meaning of a text is consistent and coherent across multiple sentences or paragraphs. It is important for understanding long or complex texts where meaning focuses on previous statements. Key aspects of discourse integration:

- Anaphora Resolution: Anaphora refers to the use of pronouns or other references that depend on earlier parts of the text. For example in the sentence "Taylor went to the store. She bought groceries" pronoun "She" refers back to "Taylor." It ensures that references like these are correctly understood by linking them to their antecedents.
- Contextual References: Many words or phrases can only be fully understood when considered in the context of following sentences. It helps in interpreting how certain words or phrases focus on context. For example "It was a great day" is clearer when you know what event or situation is being discussed.

Example of Discourse Integration

- "Taylor went to the store to buy some groceries. She realized she forgot her wallet." Understanding that "Taylor" is the antecedent of "she" is important for understanding a sentence's meaning.
- "This is unfair!" helps in understanding what "this" refers to; we need to identify the following sentences. Without context, the statement's meaning remains unclear.

It is important for NLP applications like machine translation, chatbots and text summarization. It ensures that meaning remains the same across sentences which helps machines to understand context. This enables accurate and natural responses in applications like conversational AI and document translation.

Pragmatic Analysis: Pragmatic analysis helps in understanding the deeper meaning behind words and sentences by

looking beyond their literal meanings. While semantic analysis looks at the direct meaning it considers the speaker's or writer's intentions, tone and context of the communication. Key tasks:

- **Understanding Intentions:** Sometimes language doesn't mean what it says literally. For example when someone asks "Can you pass the salt?" it's not about ability but a polite request. It helps to understand the true intention behind such expressions.
- **Figurative Meaning:** Language often uses idioms or metaphors that can't be taken literally.

Examples of Pragmatic Analysis: "Hello! What time is it?" here it might be a straightforward request for the current time but it could also imply concern about being late.

For example "I'm falling for you" means "I love you" not literally falling. It helps to interpret these non-literal meanings. It is important for NLP tasks like sentiment analysis, chatbots and conversation-based AI. It helps machines to understand the speaker's intentions, tone and context which go beyond the literal meaning of words. By identifying sarcasm and emotions this helps systems to respond naturally which improves human-computer interaction. By combining these phases NLP systems can effectively interpret, analyze and generate human language making more intelligent and natural interactions between humans and machines.

Biology of Speech Processing:

The biology of speech processing involves understanding how human brains and bodies produce and perceive speech, which forms the foundation for technologies in natural language processing (NLP) and AI-driven speech systems.

Biological Foundations of Speech Processing

Speech production is driven by the vocal cords vibrating to create sound, which is then shaped by the tongue, lips, and other articulators into recognizable phonemes (the smallest units of sound in a language).

- The human speech production system converts mental representations of language into vocal signals, while the auditory system allows us to decode sounds back into meaning.
- The brain manages complex processes such as speech segmentation (distinguishing words in a continuous stream of speech), parsing syntax, and understanding semantics, all in real-time.

Speech Processing in NLP:

Speech processing in computer science focuses on converting spoken language into text (speech recognition) and vice versa (text-to-speech synthesis). This process involves:

- Capturing and digitizing speech signals.
- Using probabilistic and machine learning models to decode these signals into phonemes, words, and sentences.
- Handling variations like accents, tone, coarticulation (blending of sounds), and noise.

Traditional methods (like Hidden Markov Models) have largely been replaced by deep neural networks and transformer models for both speech recognition and language understanding.

- Transformer-based models (e.g., BERT, GPT) use vast amounts of data to learn language context and nuance, enabling more accurate transcription and interpretation of speech and text.

NLP also encompasses tasks such as:

- Text classification (e.g., sentiment analysis, topic modeling)
- Information extraction (entities, relationships)
- Machine translation and text summarization
- Dialogue systems (chatbots, virtual assistants)
- Speech synthesis (producing realistic speech from text)

Connection Between Biology and NLP

- NLP technologies are inspired by and model biological processes: how humans segment, interpret, and generate language. Early NLP was closely linked to cognitive linguistics and psycholinguistics, although statistical and neural approaches now dominate mainstream methods.
- Cognitive models aim to mimic or explain biological language processing for improved explainability and performance in AI systems, though such models are less common.

Place and Manner of Articulation

- **Place of Articulation:** The specific point in the vocal tract where two speech organs (articulators) come together

to produce a consonant sound.

- **Manner of Articulation:** Describes how the airstream is manipulated or modified by the articulators to produce different types of speech sounds.

Manner	Description	Example Sounds	Example Words
Stops (Plosives)	Complete stop and release of airflow	/p/, /t/, /k/, /b/, /d/, /g/	"pen", "top", "go"
Fricatives	Narrow airflow creates friction	/f/, /v/, /s/, /z/, /ʃ/, /ʒ/, /h/	"fish", "zoo", "she"
Affricates	Stop + Fricative combo	/tʃ/, /dʒ/	"chip", "judge"
Nasals	Air flows through the nose	/m/, /n/, /ŋ/	"man", "no", "song"
Liquids	Smooth airflow around tongue	/l/, /r/	"lip", "red"
Glides	Slight movement like vowels	/w/, /j/	"win", "yes"

Why It Matters:

- **Phonetic Modeling:** Place and manner of articulation are core elements in phonetic models. Differentiating how and where a sound is produced enables models to recognize phonemes accurately, even in complex, noisy environments or among speakers with different accents.
- **Speech Recognition Accuracy:** Incorporating articulatory features (place, manner, voicing, aspiration) lets systems interpret subtle differences between similar sounds. This reduces errors when transcribing speech to text and improves robustness in tasks like segmentation (word boundary detection) and word identification.
- **Pronunciation and Synthesis:** To generate natural-sounding speech, as in text-to-speech synthesis, understanding how articulation affects acoustic properties is essential. Correct articulation information helps computers produce speech that's understandable and realistic.
- **Handling Linguistic Diversity:** Articulatory features help systems model the diversity of world languages, many of which share similar articulatory mechanisms. This universality streamlines multilingual processing and aids in language learning tools.
- **Speaker Identification:** Subtle articulatory differences contribute to unique voice prints, supporting speaker recognition and identification in security or personalization tasks.
- **Phoneme Classification:** By classifying speech sounds according to their place and manner, NLP systems can disambiguate words and understand pronunciation variations (allophones), helping with more precise automatic transcription and language understanding.

Application in NLP

- **Speech Recognition:** NLP systems use knowledge of place and manner of articulation to convert acoustic signals into phonemes. Recognizing how and where a sound is made helps in distinguishing between similar-sounding phonemes.
- **Phoneme Modeling:** Acoustic models in NLP and speech processing often use features based on articulatory properties, improving the models' ability to handle accents, dialects, and noisy input by simulating human perception.
- **Automatic Phonetic Transcription:** Systems can predict or check pronunciation by analyzing place and manner data, crucial for language learning or speech therapy applications.
- **Text-to-Speech (TTS):** Accurate generation of speech requires synthesizing the correct articulatory movements corresponding to place and manner, making spoken output more natural and intelligible.

Word boundary detection

Word boundary detection in NLP refers to the task of identifying where individual words begin and end within continuous speech or unsegmented text. This is particularly challenging in speech processing, as spoken language often lacks clear pauses between words and is affected by phenomena like coarticulation, varying speaking rates, and accent differences.

Techniques:

- Acoustic Cues: Utilizing changes in pitch, duration, and intensity of the speech signal to infer word boundaries.
- Phonotactic Rules: Language-specific rules about which phoneme sequences are permissible can help identify boundaries.
- Statistical Methods: Probabilistic models like Hidden Markov Models (HMMs) or neural networks can predict word boundaries based on learned patterns.
- Lexical Information: Using dictionaries and known word lists to match spoken segments to possible words.

Importance and Applications

- Speech Recognition: Accurate word boundary detection is crucial for transcribing speech into text reliably, which in turn affects keywords spotting, machine translation, and conversational AI systems.
- Language Acquisition & Zero Resource Processing: Unsupervised word boundary detection mimics how infants might learn to segment speech, and it is critical in developing language models for low-resource or unwritten languages.

Methods and Approaches

- Acoustic-based Techniques
 - Rely on features such as energy, pitch, pauses, and the distinction between voiced/unvoiced/silence segments.
 - Short Time Energy and Zero Crossing Rate (ZCR): Variations can indicate word boundaries, especially when a segment transitions from voiced/unvoiced to silence or vice versa.
 - Teager Energy Operator (TEO): Used alongside pitch and energy metrics for detecting word onsets, especially in stress-bound languages like English.
 - Pause and Prosody: Pauses and changes in speech rate and intonation can also signal boundaries, though coarticulation often complicates this in natural speech.
- Combined Acoustic and Lexical Features
 - State-of-the-art systems often combine acoustic cues (like the above) with lexical cues (e.g., known word patterns or language-specific stress rules).
 - Hidden Markov Models (HMMs) and other statistical methods are often trained to incorporate both feature sets, yielding higher boundary detection accuracy (F-score up to 0.81).

Self-Supervised and Deep Learning Methods

- Unsupervised Boundary Detection: Methods such as those leveraging self-supervised models (e.g., HuBERT features) detect dissimilarity between adjacent frames. When dissimilarity peaks (after smoothing), these are marked as probable word boundaries—reflecting shifts in acoustic characteristics likely corresponding to word changes.
- Clustering: Extracting segments using detected boundaries and grouping similar segments builds a lexicon, which aids the unsupervised discovery of words in novel languages.
- Prominence-Based Segmentation: Instead of just using hard thresholds, some newer methods use relative prominence within the dissimilarity curve to identify more nuanced boundary points.

Key Challenges

- Coarticulation: Overlapping sounds between words often obscure clear-cut acoustic boundaries.
- Speaker Variability and Noise: Differences in accent, speaking style, and background noise require robust, adaptable features and models.

Argmax Based Computations:

The argmax function returns the input (argument) at which a given function reaches its maximum value. It differs from max, which returns the maximum value of the function rather than the input that produces it.

Use in NLP and Machine Learning

- Most common in classification tasks where models output a vector of probabilities for classes.
- Argmax identifies the class with the highest predicted probability.

Specific NLP Applications

- Word or token prediction: In sequence-to-sequence models (e.g., translation, text generation), argmax selects the most likely next word/token at each step.
- Classification: Assign labels based on the highest scoring class.
- Decoding: Argmax is used in decoding schemes like greedy decoding for generating sentences.
- Parsing and tagging: Selecting the most probable parse tree or POS tag from candidate options.

Computation & Models

- Often used with models that produce probabilistic outputs or likelihoods such as Hidden Markov models (HMMs), Conditional Random Fields (CRFs), or neural networks.
- Argmax is crucial in noisy channel models and Bayesian decision theory where the goal is to maximize posterior probabilities.

Challenges and Differentiability

- Argmax is not differentiable, which creates challenges in training models end-to-end via backpropagation.
- Techniques include approximations or alternative methods (e.g., softmax followed by sampling).

Example in ML Workflow

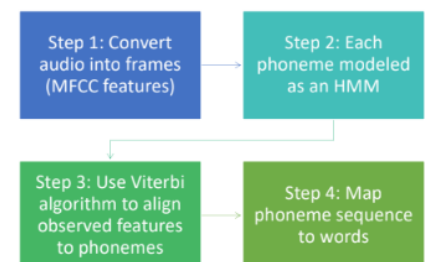
- A neural network outputs a probability distribution over vocabulary for the next word.
- At inference, argmax picks the word with the highest probability.
- Example: Vocabulary indices = {0: "cat", 1: "dog", 2: "mouse"}
- Model output probabilities: [0.2, 0.7, 0.1] → argmax = 1 → predicted word is "dog".

HMM and Speech Recognition

Definition: Hidden Markov Models (HMMs) are statistical models used to represent time-series data like speech. Each state in an HMM represents a phoneme or word and transitions between states are governed by probabilities.

Components:

- States: Represent different parts of the speech signal, such as different phonemes or words.
- Observations: The actual speech sounds (acoustic signals) that are heard.
- Transition Probabilities: The probabilities of moving from one state to another.
- Emission Probabilities: The probabilities of producing a particular observation from a state.



Speech Recognition:

Definition: The process of converting spoken language into text.

Process:

- Feature Extraction: Convert the speech signal into a sequence of feature vectors that represent the important characteristics of the sound.
- Modeling: Use HMM to model the sequence of feature vectors as a sequence of states.
- Decoding: Find the most likely sequence of words given the observed feature vectors using algorithms like the Viterbi algorithm.

Example:

- Training: Train the HMM on a labeled dataset of speech and corresponding text. This involves estimating the transition and emission probabilities from the data.
- Recognition: Use the trained HMM to recognize new speech input. Given a new sequence of feature vectors, the HMM finds the most probable sequence of states (words) that could have generated the observed sequence.

Use Case in NLP:

- Part-of-Speech Tagging
- Named Entity Recognition
- Speech Recognition